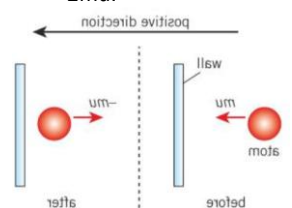
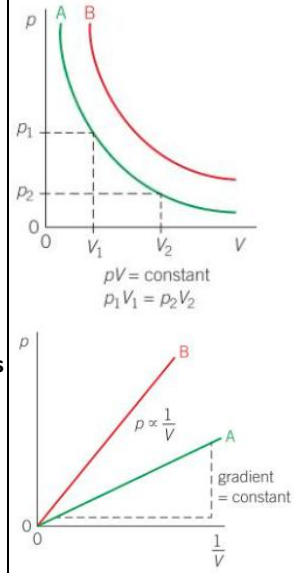


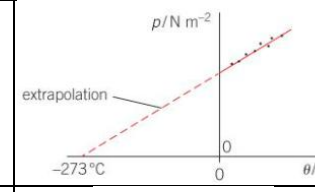
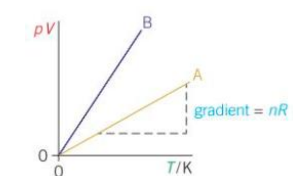
1. Lesson 1 The kinetic theory of gases

mole	The amount of substance that contains as many elementary entities as there are atoms in 0.012 kg (12 g) of carbon-12.
Avogadro constant	6.02×10^{23} , the number of atoms in 0.012 kg (12 g) of carbon-12: symbol N_A
Total number of atoms; N	$N = n \times N_A$ n – number of moles of a substance
Molar mass	The mass of one mole of a substance.
Nucleon number	The total number of protons and neutrons in a nucleus (also called the mass number); symbol A
Ideal gas	A model of a gas including assumptions that simplify the behaviour of real gases.
Kinetic theory of matter	The kinetic theory of matter is a model used to describe the behaviour of the atoms or molecules in an ideal gas.
The assumptions made in the kinetic model for an ideal gas	- The gas contains a very large number of atoms or molecules moving in random directions with random speeds. - The atoms or molecules of the gas occupy a negligible volume compared with the volume of the gas. - The collisions of atoms or molecules with each other and the container walls are perfectly elastic (no kinetic energy is lost). - The time of collisions between the atoms or molecules is negligible compared to the time between the collisions. - Electrostatic forces between atoms or molecules are negligible except during collisions.

Pressure in gases	Pressure, P , of a gas is the force, F per unit area, A . It is measured in Pascals, (Pa) where $1 \text{ Pa} = 1 \text{ Nm}^{-2}$ $p = \frac{F}{A}$
Change in momentum (applied in gases)	The atoms or molecules in a gas are always moving, and when they collide with the walls of a container the container exerts a force on them, changing their momentum as they bounce off the wall. When a single atom collides with the container wall elastically, its speed does not change, but its velocity changes. The total change in momentum is $-2mu$. 
Newton's second law (applied in gases)	The atom bounces between the container walls, making frequent collisions. The force acting on the atom is $F_{\text{atom}} = \Delta p / \Delta t$ where $\Delta p = -2mu$ Δt – the time between collisions with the wall
Newton's third law (applied in gases)	The atom exerts an equal but opposite force on the wall.

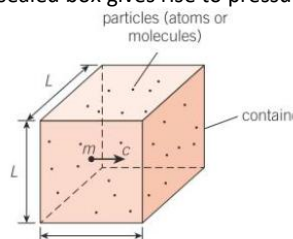
2. Lesson 2 Gas laws

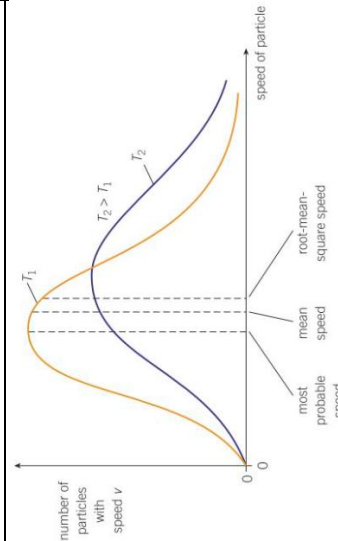
Gas laws	The laws governing the behaviour of ideal gases, like Boyle's law. The relationships between the temperature, pressure, and volume of an ideal gas can be described by gas laws.
Boyle's law (temperature constant)	The pressure of an ideal gas is inversely proportional to its volume, provided that the mass of gas and the temperature do not vary.
Pressure-volume graphs for two gases at different temperatures	
Gas Law (volume constant)	If the volume and mass of gas remains constant, the pressure p of an ideal gas is directly proportional to its absolute (thermodynamic) temperature T in kelvin. $p \propto T$ or $\frac{p}{T} = \text{constant}$

A graph of pressure of gas against its temperature	
Combining the laws	$\frac{pV}{T} = \text{constant}$ $\frac{p_{\text{initial}} V_{\text{initial}}}{T_{\text{initial}}} = \frac{p_{\text{final}} V_{\text{final}}}{T_{\text{final}}}$ or simply $\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$
Molar gas constant	The constant in the equation of state of an ideal gas – symbol R , $8.314 \text{ JK}^{-1} \text{ mol}^{-1}$
Equation of state of an ideal gas	$\frac{pV}{T} = nR \quad \text{or} \quad pV = nRT$
Graphical analysis	A graph of pV against T for a fixed amount of gas produces a straight line through the origin. $pV \propto T$  The greater the number of moles of gas, the steeper the line becomes.

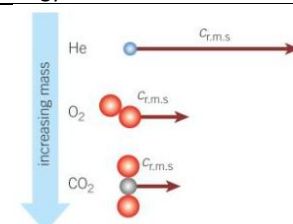
3. Lesson 3 Root mean square speed

Root mean square speed	The square root of the mean square speed (of all the particles in a gas). $\sqrt{c^2} \quad \text{or} \quad c_{\text{r.m.s}}$
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Root mean square speed $/c_{r.m.s.}$	- Provides a useful way to describe the motion of the particles in a gas and it is an average.
Mean square speed	The mean of the squared velocities (of all the particles in a gas). $\overline{c^2}$
Pressure at the microscopic level	The equation for the pressure and volume of a gas: $pV = \frac{1}{3} N m \overline{c^2}$ p – pressure exerted by the gas V – volume of the gas N – is the number of particles in the gas m – is the mass of each particle $\overline{c^2}$ – is the mean square speed of the particles The equation above can be derived by considering how the movement of atoms or molecules of gas inside a sealed box gives rise to pressure. 
Maxwell – Boltzmann distribution	The distribution of the speeds of particles in a gas.

	
Graph description	Changing the temperature of the gas changes the distribution. The hotter the gas becomes, the greater the range of speeds.
4. Lesson 4 The Boltzmann constant	
Ludwig Boltzmann	Austrian physicist, whose greatest achievement was arguably his work on statistical mechanics.
Boltzmann constant, k	The molar gas constant R divided by the Avogadro constant N_A, a constant that relates the mean kinetic energy of the atoms or molecules in a gas to the gas temperature – symbol k. $k = \frac{R}{N_A} = 1.38 \times 10^{-23} \text{ J K}^{-1}$

Ideal gas equation	You can substitute the definition of k into the ideal gas equation $pV = nRT$ to give $pV = nkN_A T$ the number of particles in the gas sample, N, is equal to $n \times N_A$ therefore $pV = nRT$
Mean kinetic energy and temperature	By combining $pV = \frac{1}{3} N m \overline{c^2}$ and $pV = nRT$ we can derive an expression which directly relates the mean kinetic energy of particles in a gas to the absolute temperature of the gas $\frac{1}{2} m \overline{c^2} = \frac{3}{2} kT$
Mean average kinetic energy of the particles in the gas	$\frac{1}{2} m \overline{c^2}$ This only applies if the temperature is measured in kelvin.

Particle speeds at different temperatures	- At a given temperature the atoms or molecules in different gases have the same average/mean kinetic energy. - as the particles have different masses their r.m.s. speeds will be different. - The oxygen molecules and helium atoms around you, in spite of their different masses have the same mean kinetic energy.
The r.m.s. speed of different particles	
The internal energy of an ideal gas	- The internal energy of a gas is the sum of the kinetic and potential energies of the particles inside the gas. - The internal energy of an ideal gas is the sum of the kinetic energy of the particles. - One of the assumptions of an ideal gas states that the electrostatic forces between particles in the gas are negligible except during collisions.